

Emir Chacra

Ph.D. Candidate in Computer Music

www.emirchacra.com

ABOUT

<i>Full Name</i>	Emir Nicolás Chacra
<i>Birth</i>	1st of October, 1996
<i>Nationality</i>	Chilean
<i>City</i>	San Diego, California
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EDUCATION

UC San Diego, Ph.D. in Computer Music ABD in May 2026. Advisor: Tom Erbe	09/23 - Ongoing
Universidad de Chile, M.S. in Applied Mathematics Grade: 6.8 out of 7.0 (GPA of 4.0). Graduated with honors. Dissertation: "Optimization of the anisotropic elastic properties of a mechanical metamaterial for the design of wooden musical instruments". Advisor: Axel Osses.	03/21 - 05/22
Universidad de Chile, Mathematical Engineering Grade: 7.0 out of 7.0 (GPA of 4.0). Graduated with honors.	03/15 - 05/22

PUBLICATIONS

Peer-reviewed journal articles

1. A. Osses, P. Smith*, R. Calvo, E. Blanco, **E. Chacra**, E. M. Gayo, C. Amigo, A. Urquiza. A conceptual model for the analysis of heat impacts on indoor thermal comfort and urban climate resilience: application to medium-sized cities in Chile. Accepted in Sustainable Cities and Society. (*corresponding author)
2. S. Gonzalez, **E. Chacra**, C. Carreño, C. Espinoza, "Wooden Mechanical Metamaterials: towards tunable wood plates", Materials and Design, 221, 110952, 2022.

Conference proceedings

3. Sergey Kasich, and **Emir Chacra**. 2026. Knotty Oscillator: Breaking knot topology for a new physically-inspired sound generator. Proceedings of the International Conference on New Interfaces for Musical Expression. DOI: 10.5281/zenodo.20784090
4. **E. Chacra**, P. Escapil-Inchauspé, L. Martí and N. Sanchez-Pi, "Physics-Informed Neural Networks for Coupled Fluid Dynamics and Nutrient Transport: A Comparative Study of Causal Training Methods," 2025 15th IEEE International Conference on Pattern Recognition Systems (ICPRS), Valparaiso, Viña del Mar, Chile, 2025, pp. 1-7.
5. C. Espinoza, C. Carreño, **E. Chacra** and S. Gonzalez, "MetaWood: manipulation of the elastic properties of wood plates by periodic hole patterns", 2022 Sixteenth International Congress on Artificial Materials for Novel Wave Phenomena (Metamaterials), 2022, pp. 139-141.

TEACHING EXPERIENCE

As an instructor

- UCSD/MACSD Audio Workshops: Beatmaking. (07/24 - 09/24)

As a guest speaker

- Talk and workshop: *Diseño y programación de efectos de audio basados en delay..* Sound Department, Universidad de Chile. Dec 2025.
- Talk and workshop: *What should I do to start designing software or hardware digital audio effects?.* Sound Department, Universidad de Chile. Dec 2024.

As a teaching assistant

- UC San Diego:
 - MUS 174C - Audio/MIDI Studio Techniques III. Spring 2026 (1 quarter).
 - MUS 174B - Audio/MIDI Studio Techniques II. Winter 2026 (1 quarter).
 - MUS 8 - American Music : Jazz Cultures. Fall 2025 (1 quarter).
 - MUS 172 - Computer Music II. Spring 2024, Spring 2025 (2 quarter).
 - MUS 171 - Computer Music I. Fall 2023, Winter 2025 (2 quarters).
 - MUS 170 - Art of Sound. Fall 2024 (1 quarter).
 - MUS 17 - Hip Hop. Winter 2024 (1 quarter).
- Universidad de Chile:
 - Probability and statistics. Fall 2020, Fall 2022, Spring 2022 (3 terms).
 - Numerical analysis of partial differential equations. Fall 2022, Fall 2021 (2 terms).
 - Probability and statistics in Data Analysis. Fall 2019 (1 term).
 - Statistics. Spring 2018 (1 term).

RESEARCH EXPERIENCE

INRIA Chile, R&D Engineer Advisor: Hugo Carrillo.	03/23 - 08/23
CMM, Universidad de Chile, Research Assistant Advisor: Axel Osses.	07/22 - 07/23
CMM, Universidad de Chile, Research Assistant Advisors: Jocelyn Dunstan, Alejandro Maass.	01/21 - 03/21
ALeRCE, Universidad de Chile, Research Assistant Advisor: Francisco Förster.	04/19 - 12/19

MUSIC PRODUCER EXPERIENCE

early riser - monocromo (EP) Producer as <i>monocromo</i> and mixing engineer as <i>Emir Chacra</i> .	11/24
Panoramas - monocromo (LP) Producer as <i>monocromo</i> and mixing engineer as <i>Emir Chacra</i> .	06/23
Ambientes - monocromo (EP) Producer as <i>monocromo</i> and mixing engineer as <i>Emir Chacra</i> .	11/22
Neonatología - Nuevalengua (LP) Co-Producer as <i>Emir Chacra</i> .	08/18

BASSIST EXPERIENCE

monocromo

01/22 - Ongoing

Recordings: Ambientes EP (Nov 2022).

Nuevalengua

05/16 - 09/18

Recordings: Neonatología LP (Aug 2018).

Live performances: Converse Bold Park (2016), Festival Walén (2016), Nuevalengua at Goodstock Bar (2017), Nuevalengua at “La Casa del Sol” (2018).

PERSONAL SKILLS

Programming Python (Advanced), C++ (Moderate), Git (Moderate), Matlab (Moderate).

Audio software Pd (Advanced), Bitwig (Advanced), Ableton Live (Advanced), Max/MSP (Moderate).

Languages Spanish (Native proficiency), English (Full professional proficiency), Portuguese (Elementary proficiency).